

***Example**

Sketch the root locus of the system if $G(s)H(s) = \frac{K}{s(s^3+12s^2+64s+128)}$

Step 1:

Write the characteristic equation in the form $1 + G(s)H(s) = 0$; $1 + \frac{K}{s(s^3+12s^2+64s+128)} = 0$

Step 2:

Determine the poles and zeros of $G(s)H(s)$. The poles are $s = 0, -4, -4 \pm j4$. There are no zeros. The number of poles (n) = 4.

The number of zeros (m) = 0.

The root loci originate from the poles of $G(s)H(s)$ and end at the zeros.

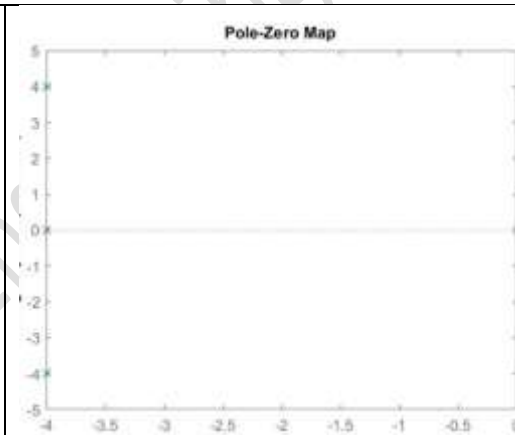
The number of branches of root loci proceeding to the zeros at infinity = $n - m = 4$

Step 3:

Number of separate loci is equal to the number of poles (n)

Step 4:

Locate the segments of real axis that are part of the root loci. The real axis segment to the left of odd number of poles and zeros will be part of the root loci. The first pole/zero is located at the origin. So the part of real axis from $s = 0$ till the next pole/zero at $s = -4$ will be part of the loci. Since there are no more poles/zeros in the real axis after $s = -4$, no more part of real axis will form a part of root loci.



Step 5:

The root loci are symmetrical about the real axis

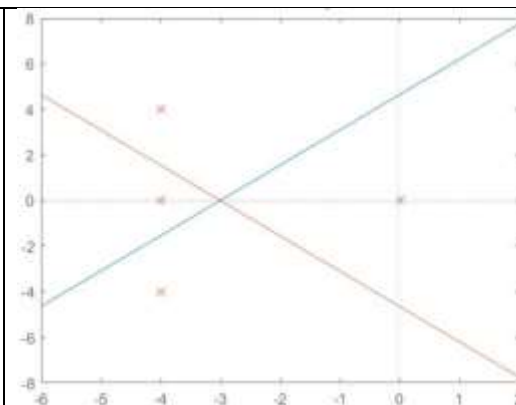
Step 6:

The root loci proceed to the zeros at infinity along the asymptotes centered at

$$\sigma_A = \frac{\sum \text{poles} - \sum \text{zeros}}{n-m} = \frac{(0-4-4+j4-j4)-(0)}{4-0} = \frac{-12}{4} = -3$$

$$\phi_A = \frac{(2k+1)180^\circ}{n-m}; k = 0, 1, 2, 3$$

$$\phi_A = \frac{180^\circ}{4}, \frac{3 \times 180^\circ}{4}, \frac{5 \times 180^\circ}{4}, \frac{7 \times 180^\circ}{4} = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$



Step 7:

Determine the break away/break in point.

Write the characteristic equation $K = P(s)$

Determine the roots of $\frac{dK}{ds} = 0$

$$1 + \frac{K}{s(s^3+12s^2+64s+128)} = 0$$

$$K = -s(s^3 + 12s^2 + 64s + 128)$$

$$\frac{dK}{ds} = -(4s^3 + 36s^2 + 128s + 128) = 0$$

$$s = -3.7117 \pm j2.5533, -1.5767$$

We take the value $s = -1.57$ for the breakaway point

Step 8:

Determine the imaginary axis crossing by constructing the Routh table.

The characteristic equation is $1 + \frac{K}{s(s^3 + 12s^2 + 64s + 128)} = 0$

$$s^4 + 12s^3 + 64s^2 + 128s + K = 0$$

s^4	1	64	K
s^3	12	128	
s^2	53.33	K	
s^1	$\frac{53.33 \times 128 - 12 \times K}{53.33}$		
s^0	K		

$$\frac{12 \times 64 - 1 \times 128}{12} = 53.33$$

$$\frac{12 \times K - 1 \times 0}{12} = K$$

$$\frac{53.33 \times 128 - 12 \times K}{53.33}$$

When the root loci cross the imaginary axis, there will be a row of zeros. This is possible only in the s^1 row when $53.33 \times 128 - 12 \times K = 0$ or $K = 568.89$.

The auxiliary polynomial is $53.33s^2 + 568.89 = 0$

Or $s = \pm j3.26$. is the point where the root loci crosses the imaginary axis

Step 9:

Determination of the angle of locus departure from the complex poles and angle of locus arrival at complex zeros from the angle criterion.

Consider a test point s very close to the complex pole at $-4 + j4$. Let θ_1 be the angle subtended by the vector from the pole at $-4 + j4$ to the test point s

θ_2 from -4 to $s = 90^\circ$

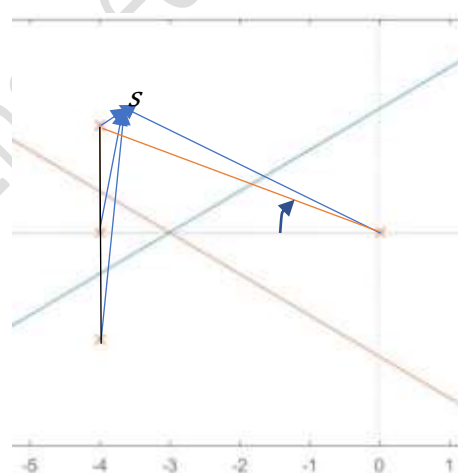
θ_3 from $-4 - j4$ to $s = 90^\circ$

θ_4 from 0 to $s = 135^\circ$ ($\tan^{-1} \frac{4}{4} = 45^\circ$)

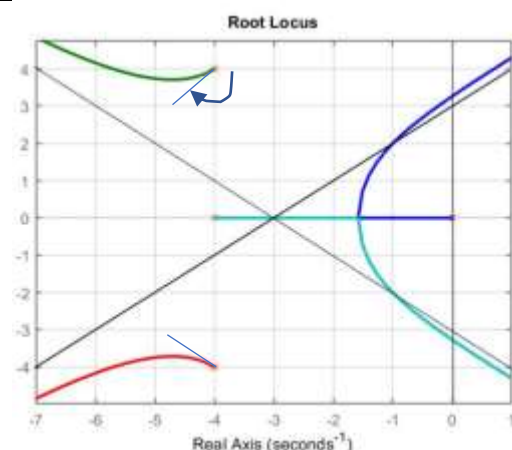
$\theta_1 + \theta_2 + \theta_3 + \theta_4 = \text{odd multiple of } 180^\circ$

$$\theta_1 = 180 - 90 - 90 - 135 = -135^\circ$$

Since the root loci is symmetrical about the real axis, the test point s in the vicinity of the pole at $-4 - j4$ will have an angle contribution of $+135^\circ$

**Step 10:**

Determine the root locations which satisfy the angle criterion and join such points to get the locus.



#Exercise 36

Sketch the root locus of the system if $G(s)H(s) = \frac{K}{s(s+4)(s^2+2s+2)}$. Verify your answer by plotting it in Matlab/Scilab.

#Exercise 37

Sketch the root locus of the system if $G(s)H(s) = \frac{K(s+6)}{s(s+3)(s^2+2s+2)}$. Verify your answer by plotting it in Matlab/Scilab.

#Exercise 38

Sketch the root locus of the system if $G(s)H(s) = \frac{K(s+4)(s+6)}{s(s+2)}$. Verify your answer by plotting it in Matlab/Scilab.

#Exercise 39

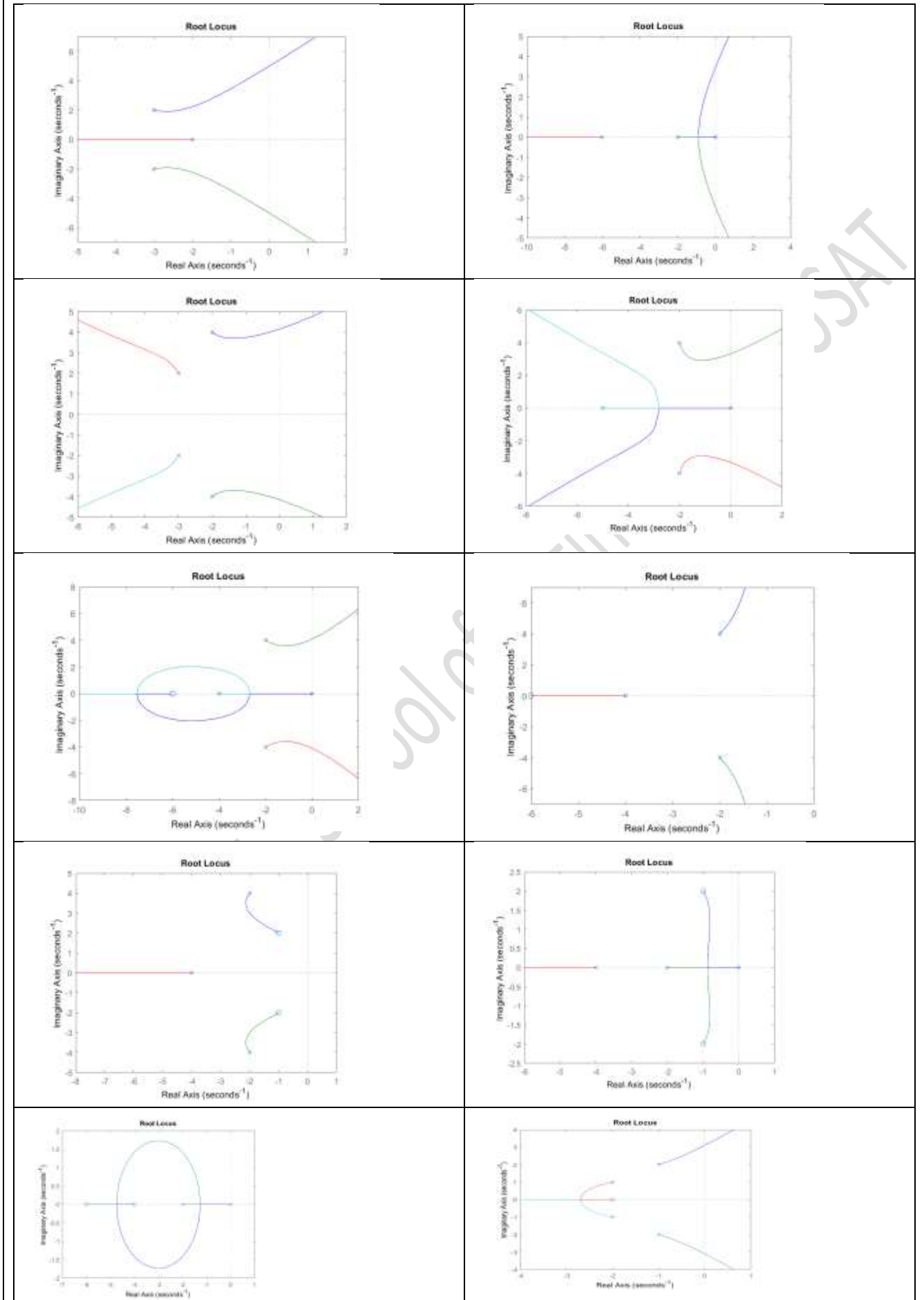
Sketch the root locus of the system if $G(s)H(s) = \frac{K}{s(s+3)(s+5)(s^2+2s+2)}$. Verify your answer by plotting it in Matlab/Scilab.

#Exercise 40

Using Matlab/Scilab, sketch the root locus of the system whose $G(s)H(s) = \frac{K}{s(s+5)}$. Observe the changes in the root locus when it is modified by adding poles and zeros as given below

- i) $G(s)H(s) = \frac{K}{s(s+5)(s+10)}$
- ii) $G(s)H(s) = \frac{K(s+1)}{s(s+5)}$
- iii) $G(s)H(s) = \frac{K(s+10)}{s(s+5)}$
- iv) $G(s)H(s) = \frac{K(s+1)}{s(s+5)(s+10)}$
- v) $G(s)H(s) = \frac{K(s+1)(s+10)}{s(s+5)(s+3)}$

Open loop poles and zeros and the corresponding root loci



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